CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M. Sc. (Microbiology) Examination, March 2019

Course code & Name: MI 823 & Environmental Microbiology

Date: 19/03/2019 Day: Tuesday Time: 1:30 P.M. To 2:00 P.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	**Choose the correct answer for the following questions.**		**20**
1.	The actinomycete *Pseudonocardia* produces an antimicrobial compound that inhibits the growth of *Escovopsis* this relationship can be classified as _____________.		01
	(i) Predation	(ii) Parasitism	
	(iii) Competition	(iv) Amensalism	
2.	A lichen is formed by mutualism between an algae and ________.		01
	(i) Diatom	(ii) Fungus	
	(iii) Phytoplankton	(iv) Cyanobacteria	
3.	The following method of microbial diversity analysis employs a gradient of a DNA denaturant to separate rRNA genes of the same size that differ in their melting profile.		01
	(i) DGGE	(ii) ARISA	
	(iii) FAME	(iv) Phylochip	
4.	The following marine phytoplanktonic Cyanobacteria are the numerically dominant photosynthetic cells on the planet.		01
	(i) *Oscillatoria*	(ii) *Phormidium*	
	(iii) *Pelagibacter*	(iv) *Prochlorococcus*	

5.	The following is a measure of species diversity in a sample unit and is often determined using species richness or a diversity index.		01
	(i) Beta (β) diversity	(ii) Alpha (α) diversity	
	(iii) Gamma (γ) diversity	(iv) All of them	
6.	The following filamentous fungus is responsible for sludge bulking in the activated sludge process.		01
	(i) *Enterococcus aerogenes*	(ii) *Geotrichum candidum*	
	(iii) *Sphaerotilus natans*	(iv) *Cyrptococcus neoformans*	
7.	The bacterium *Helicobacter pylori*, produces large amounts of the following enzyme to generate ammonia for neutralization of gastric hydrochloric acid in human stomach.		01
	(i) Carbonic anhydrase	(ii) Peptidase	
	(iii) Lipase	(iv) Urease	
8.	Phylogenetic FISH stains used to observe prokaryotic microorganisms are fluorescent-tagged oligonucleotides probes complementary to _________.		01
	(i) 18S rRNA	(ii) 28S rRNA	
	(iii) 16S rRNA	(iv) All of them	
9.	In the following FISH method the nucleic acid probe is conjugated with peroxidase enzyme molecule, instead of a fluorescent dye.		01
	(i) PhyloChip	(ii) CARD-FISH	
	(iii) FISH-MAR	(iv) All of them	
10.	The following protein binds to oxygen and helps in protection of oxygen-sensitive nitrogenase enzyme in the mature root nodules.		01
	(i) Rhicadhesin	(ii) Nod factor	
	(iii) Leghemoglobin	(iv) Muropeptide	
11.	Selective isolation of single-cells of microorganisms is possible using which of the following cultivation-dependent technique.		01
	(i) Flow cytometer	(ii) T-RFLP	
	(iii) Laser tweezers	(iv) Both (i) and (iii)	
12.	In the gastro-intestinal tract of a ruminant animal, microbial fermentation generates ________, which are utilized by the animal as its main source of energy.		01
	(i) Opines	(ii) Oligosaccharides	
	(iii) Volatile fatty acids (VFAs)	(iv) Amino acids	
13.	The following compound formed by methylation of mercury is volatilized in environment.		01
	(i) Mercury	(ii) Organomercurials	
	(iii) Mercuric chloride	(iv) All of them	
14.	The following molecule protects microbes from metal toxicity by sequestering metal ions.		01
	(i) Metallothioneins	(ii) MerT protein	
	(iii) CadA protein	(iv) None of them	
15.	A microorganism producing the following compound is more suitable for bioremediation of metals, as it helps to mobilize bound metal ions.		01
	(i) Biosurfactant	(ii) Biopesticide	

	(iii) Siderophore	(iv) Biostimulant	
16.	The following is enzyme(s) is/are involved in lignin modification		01
	(i) Laccase	(ii) LiP	
	(iii) MnP	(iv) All of them	
17.	Introduction of air into the saturated zone below the water table for bioremediation of ground water is called		01
	(i) Bioaugmentation	(ii) Biosparging	
	(iii) Bioventing	(iv) Aeration	
18.	The following term describes the chemical breakdown of a substance to smaller molecules caused by microbes or enzymes.		01
	(i) Bioaccumulation	(ii) Mineralization	
	(iii) Biodegradation	(iv) Biotransformation	
19.	The following process allows complete oxidation of solid waste.		01
	(i) Pyrolysis	(ii) Gasification	
	(iii) Incineration	(iv) All of them	
20.	Which of the following pathway degrades aliphatic hydrocarbons.		01
	(i) Glycolysis	(ii) TCA	
	(iii) β- oxidation	(iv) Catechol	

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III Semester of M. Sc. (Microbiology) Examination, March 2019

Course code & Name: MI 823 & Environmental Microbiology

Date: 19/03/2019 Day: Tuesday Time: 2:00 P.M. To 4:30 P.M. Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q – II Answer the following questions as directed (answer any five).		**20**
1.	Explain the formation of microbial biofilms and give any two examples of microbial biofilms.	04
2.	Explain the laser-tweezers technique for physically isolating individual cells for subsequent growth in pure cultures.	04
3.	Explain Nitrogen-fixing symbiosis of *Rhizobium* with leguminous plants.	04
4.	Explain the DGGE method to study microbial communities in environmental samples.	04
5.	Explain the ecological organization of microbial community using an example.	04
6.	Explain the following microbe-microbe interactions giving suitable examples: (i) Predation (ii) Amensalism	04
7.	Explain the concept of microbial habitat and niche.	04
SECTION - II		
Q-III (A) Answer briefly (answer any four).		**12**
1.	What are bioaerosols? Describe an impaction-based device for sampling bioaerosols.	03
2.	Describe the primary and secondary processes in wastewater treatment.	03
3.	Describe the different measures used to study microbial diversity.	03
4.	Describe the anaerobic digester processes for treatment of sewage sludge.	03
5.	Describe briefly different types of soil sampling devices.	03
Q-III (B) Write a short note on the following (answer any four).		**08**
1.	Environmental biosensors.	02
2.	Prepared Bed Reactor for bioremediation of contaminated soil	02
3.	Various compounds formed during biotransformation of mercury in soil	02
4.	The role of monooxygenases in biodegradation of xenobiotic pollutants	02
5.	Abiotic and biotic factors affecting the chemical speciation and bioavailability of metals in the environment.	02
Q-III (C) Discuss briefly (answer any two).		**04**
1.	In situ biorestoration of ground water.	02
2.	Role of rhizosphere microflora in phytoremediation.	02
3.	Succession of microbial community during composting.	02
Q-III (D) Discuss briefly (answer any one).		**04**

1.	Transport mechanisms involved for biodegradation of hydrocarbon under aerobic and anaerobic conditions.	04
2.	Intracellular metal resistance mechanisms in microbes.	04
Q-III (E) Justify the following statement.		**02**
	High vapour pressure should be avoided while bioventing for the bio restoration of vedose region.	